Tuning Hashes

Selected Projects in Performance Engineering of
Hash-Based Data Structures and Hash Functions

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Master of Science in Computer Science

1 Introduction

The projects in this portfolio are my greatest accomplishments from my masters' studies. Unlike my published research, which developed within an established advising relationship and with relevant coursework as a foundation, this work grew out of entirely self-directed study. I acquired the necessary background, conducted literature reviews, then defined and pursued research questions independently. These three projects are the culmination of that process. Each addresses a distinct problem in performance engineering of hash-based data structures and hash functions.

The first generalizes binary fuse filters, a recent approximate set membership structure, to support non-integer arity, the number of hash functions per query, natively. Binary fuse filters offer only a discrete tradeoff: increasing arity reduces memory overhead but increases query time. The standard approach to achieve a continuous tradeoff is to partition data between disjoint filters of different arity. Native non-integer arity make this tradeoff principled within a single construction and modestly reduces memory overhead over the partitioning approach.

The second introduces the first optimal parameter tuning algorithm for secure learned Bloom filters, a recent approximate set membership structure that combines a learned model with two secure filters. Since standard tuning algorithms do not apply, the recommended approach is simply to allocate memory evenly between the two filters. This is a poor default: it cannot target an expected false positive rate, nor bound the worst-case rate, and can waste memory when either constraint is defined. I give the first exact tuning algorithm to minimize memory footprint subject to expected and maximum false positive rate constraints, while maintaining security guarantees.

The third extends entropy-aware hash function specialization, a technique that learns which bytes of a key carry the most information and hashes only those. Existing work assumes uniform key frequency and a generic hash table model, and does not generate efficient x64 implementations. I generalize the approach with new cost models that account for non-uniform key frequency and the specific structure of the target data structure, and add code generation targeting x64. This reduces both hash latency and hash table query latency.

Together these projects move between theory and efficient implementation, driven by an experimental approach: data suggested new theory, and that theory motivated new experiments. Each required building fluency in an unfamiliar area, and each left open problems I continue to pursue. They are the clearest evidence that I have become a computer scientist.

2 Non-Uniform Binary Fuse Filters

This project was completed for CSC 592: Algorithms for Big Data.

2.1 Motivation

Distributed systems often require fast and small set data structures. For example, content delivery networks avoid caching one-hit wonders by querying a visited set [8]. Supporting such exact membership queries on a large universe[1] U requires $\Omega(\lg |U|)$ bits per key. For large enough sets, this is impractical. By permitting false positive results with some probability ϵ, the lower bound is reduced to $\lg \frac{1}{\epsilon}$ bits per key [2]. In other words, *approximate* membership queries can be much more memory efficient than exact membership queries [1, 6, 5].

A binary fuse filter [5] is a recent static probabilistic data structure for approximate membership queries. Membership information is stored in an array of m-bit values partitioned into disjoint segments. To query the filter, a key is hashed to a fingerprint value and to three array entries located in consecutive segments (see Figure 1). If the bitwise-XOR of the entries is equal to the fingerprint, then the query *might* be in the set. Otherwise, the query *definitely* is not in the set.

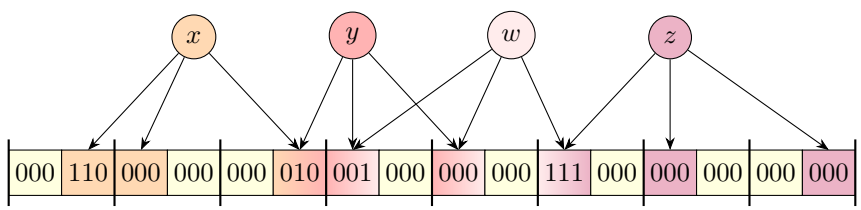

Figure 1: *Binary fuse filter with 16-entry array, 2-entry segments, and 3-bit fingerprints. Vertical bars mark segment boundaries. The filter contains keys* x, y, w *and* z, *with fingerprints* 100, 011, 110 *and* 111, *respectively. Each key hashes to three consecutive segments. Corresponding entries are marked by arrows and matching colors. For example, since the filter contains* x, *the bitwise-XOR* $110 \oplus 000 \oplus 010$ *is the fingerprint* 100.

To construct a binary fuse filter, choose each array entry such that, for each key, the bitwise-XOR of the three corresponding array entries is equal to the fingerprint. This requires solving a system of linear equations with variables in $\mathbb{F}_2^m$ (i.e. m-bit vectors) where addition is performed using bitwise-XOR. Let h_1, h_2 and h_3 be the hash functions, f be the fingerprint function, and A be the array. Each key k yields a linear equation

$$A[h_1(k)] \oplus A[h_2(k)] \oplus A[h_3(k)] = f(k)$$

[1] Set of all keys.

that must be satisfied (see Figure 2). If some equation is not satisfied, the filter will incorrectly report the corresponding key is *definitely* not in the set. Since approximate membership queries allow only false positives and not false negatives, the choice of array entries must satisfy every equation.

$$A[1] \oplus A[2] \oplus A[5] = 100 \qquad A[5] \oplus A[6] \oplus A[8] = 011$$

$$A[6] \oplus A[8] \oplus A[10] = 110 \qquad A[10] \oplus A[12] \oplus A[15] = 111$$

Figure 2: *System of linear equations with variables in $\mathbb{F}_2^3$ induced by Figure 1. Each $A[i]$ is the i-th entry in the backing array A (zero-indexed). The keys are x, y, w and z with fingerprints 100, 011, 110 and 111, respectively. Array entries and fingerprints are marked with corresponding key color(s). There is one equation for each key.*

To solve the system, model it as a hypergraph. A hypergraph is a graph where edges may contain any subset of vertices. For each entry, create a vertex. For each equation, create an edge containing entries (vertices) in the equation (see Figure 3). Then, 2-peel the hypergraph, that is:

1. Remove vertices of degree zero, then set their entries to 0^m.
2. Remove vertices of degree one and their incident edges. Then, set their entries to the bitwise-XOR of the fingerprint and the two other entries in the edge. If the other entries are not yet set, defer until they are set.
3. Repeat until there are no vertices of degree less than two, or until the empty hypergraph is reached (i.e. all entries set).

If the empty hypergraph is reached, the hypergraph is called 2-peelable. The system is solved if and only if its hypergraph is 2-peelable.

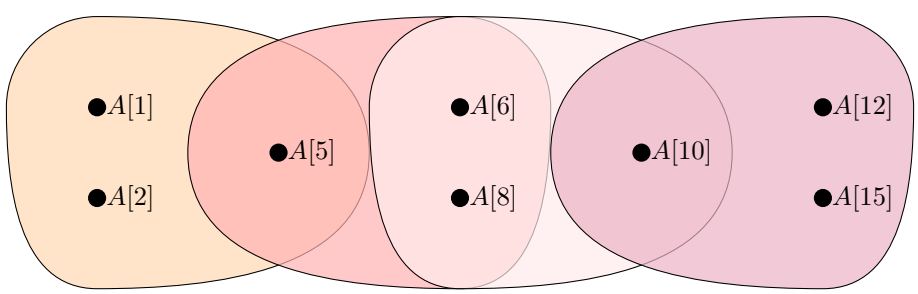

Figure 3: *Hypergraph model of Figure 2. Each $A[i]$ is the i-th entry in the backing array A (zero-indexed). Hyperedges correspond to keys x, y, w and z, colored accordingly. Vertices of degree zero not shown.*

In the filter setting, edges represent keys and vertices represent array entries. Lower edge size means cheaper queries, since fewer array entries (vertices) are accessed per key (edge). Higher edge density means lower space usage, since there are more keys (edges) per array entry (vertex).

Many hypergraph families exhibit sharp phase transitions in peelability [3, 9]. When the edge density is below a critical value, called the 2-peelability threshold, sampled hypergraphs are almost surely 2-peelable. When the density exceeds this threshold, they are almost surely not. The threshold is therefore a direct measure of space efficiency. A higher threshold means more keys can be stored per array entry.

Binary fuse filters induce the fuse graph family, which achieves the supremum 2-peelability threshold among 3-uniform hypergraph[2] families [9]. This threshold occurs at an edge density of 0.918, or roughly $1.089 \lg \frac{1}{\epsilon}$ bits per key [9]. No family of 3-uniform hypergraphs has higher 2-peelability threshold. More generally, a binary fuse filter with $m \geq 3$ hash functions[3] saturates the 2-peelability threshold for m-uniform hypergraphs [9].

However, this optimality result is confined to uniform hypergraphs. There may exist families of non-uniform hypergraphs with higher 2-peelability thresholds. If such families also have a low average edge size (i.e. few hashes and accesses per query), filters based on them could potentially reduce memory overhead without degrading query latency and throughput. Therefore, we ask:

1. How can we systematically construct non-uniform hypergraph families with peelability thresholds comparable to or surpassing those of fuse graph families?

2. How can we design efficient approximate membership query data structures by exploiting peelable non-uniform hypergraph families?

2.2 Summary

2.2.1 Non-Uniform Hypergraphs

We computed 2-peelability thresholds for 10K non-uniform fuse graph families with mean edge size between 3 and 4 (see Figure 4). As peelability is monotonic in edge density, we used binary search. At each candidate threshold, we sampled three hypergraphs with 1M edges. If the majority were peelable, we considered the family peelable at that threshold. We implemented this algorithm in C++ and found thresholds in parallel on a compute cluster.

[2] In general, a hypergraph is m-uniform if every edge contains exactly m vertices.

[3] Only binary fuse filters with three or four hash functions are practical. With four hash functions, the 2-peelability threshold is already 0.977, yielding about $1.023 \lg \frac{1}{\epsilon}$ bits per key or within 2.3% of optimal [9].

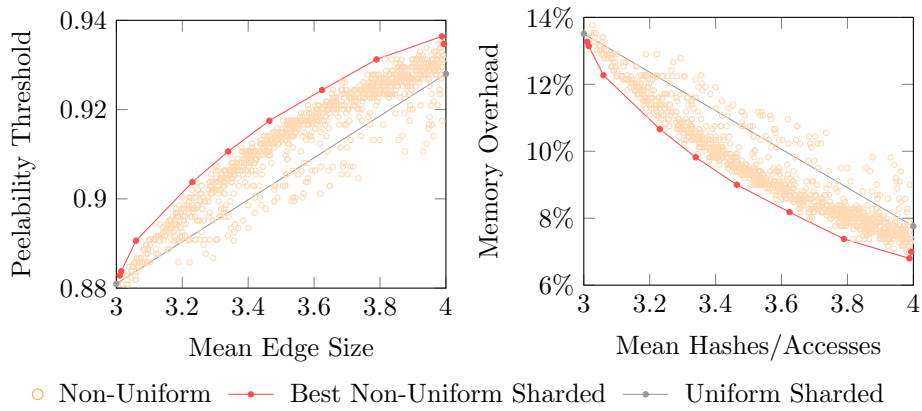

Figure 4: *Empirical peelability thresholds and memory overheads for non-uniform fuse graphs with 1M edges and mean edge size between 3 and 4. Orange dots show 1K sampled families. Red shows the best non-uniform sharded families. Gray shows the sharded uniform baseline. Many non-uniform families outperform the uniform baseline. Left: mean edge size vs. peelability threshold (higher is better). Right: theoretical query cost vs. memory overhead (lower is better).*

While m-uniform graphs have integer mean edge sizes, they can be extended to any mean edge size via sharding[4] [5]. For example, splitting edges evenly between edge-disjoint 3-uniform and 4-uniform subgraphs yields a mean edge size of 3.5. Many non-uniform families outperform this baseline, and therefore afford smaller filters. Interestingly, some non-uniform families also outperform uniform families at integer mean edge sizes. Since peelability thresholds are defined in the limit of large graphs, a non-uniform family can be more peelable in practice than a uniform family with nearly the same mean edge size. It's also possible three samples per candidate is not enough to accurately determine the threshold. For example, one family achieves $\approx 1\%$ memory savings over the uniform baseline at mean edge size 4.

These data suggest two conjectures. Let $\mathcal{H}_d$ denote the set of fuse graph families with mean edge size d and let $\tau(\mathcal{F})$ denote the 2-peelability threshold of a family $\mathcal{F}$. Define $\Gamma(d) = \sup\{\tau(\mathcal{F}) : \mathcal{F} \in \mathcal{H}_d\}$. We conjecture that $\Gamma(d)$ is continuous, monotonically increasing, and concave. Since a concave Γ would lie above any linear interpolation, this implies that sharding is suboptimal. We further conjecture that for $d \notin \mathbb{Z}$, the supremum is attained by the family with a fraction $(1-p)$ of $\lfloor d \rfloor$-edges and a fraction p of $\lceil d \rceil$-edges, where $p = d - \lfloor d \rfloor$. For example, for $d = 3.5$, this family contains 50% 3-edges and 50% 4-edges. Both conjectures remain open.

[4]Sharding partitions a dataset into disjoint subsets. For hypergraphs, each subset becomes an edge-disjoint subgraph. For filters, each subset is stored in a separate filter.

2.2.2 Non-Uniform Binary Fuse Filters

We implemented and optimized a non-uniform binary fuse filter with 50% 3-edges and 50% 4-edges in C++ (i.e. each key hashes to either 3 or 4 array entries). This edge distribution offered the greatest advantage over sharding uniform binary fuse filters. We considered three query algorithms: conditional branch, branchless and phantom entry described below (see Table 1 for metrics and Section A.2 for pseudocode).

Algorithm	Hashes	Branch Mispredictions	Cache Misses
Conditional Branch	3.5	0.5	3.5
Branchless	4	0	4
Phantom Entry	4	0	3.5

Table 1: *Metrics for non-uniform binary fuse filter query algorithms. Hashes, branch mispredictions, and cache misses are mean per query.*

In all three algorithms, we report that the key *might* be in the set if the XOR of the accessed array entries equals the fingerprint. The algorithms differ only in how they handle variable edge size when computing the XOR.

In the conditional branch algorithm (see Figure 1), we XOR the first three array entries. If the key is a 4-edge, we XOR in the fourth entry. This branch mispredicts on at least half of all queries.

In the branchless algorithm (see Figure 2), we load all four array entries unconditionally. If the key is a 3-edge, we zero the fourth value with a mask before XORing all four values. For large filters, all four loads will miss the cache.

In the phantom entry algorithm (see Figure 3), we load all four array entries unconditionally. However, if the key is a 3-edge, we zero the fourth hash address before the load, redirecting it to $A[0]$. To support this, we shift all valid entries left by one index and set $A[0] = 0$ at construction time (see Figure 5). Unlike the branchless algorithm, the fourth load for 3-edges hits the cache, since $A[0]$ is accessed in 50% of queries.

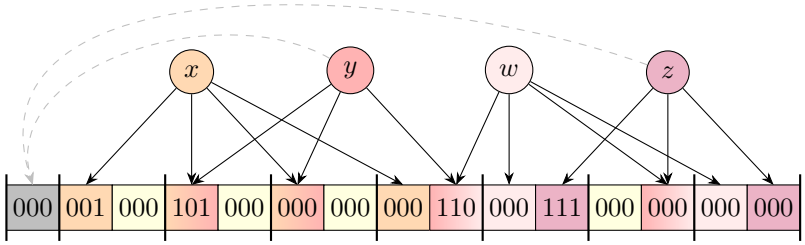

Figure 5: *Non-uniform binary fuse filter with phantom entry, 14-entry array, 2-entry segments, and 3-bit fingerprints. Vertical bars mark segment boundaries. The phantom entry is colored gray. The filter contains keys* x, y, w *and* z, *with fingerprints* 100, 011, 110 *and* 111, *respectively. Each key hashes to three or four consecutive segments. Corresponding entries are marked by arrows and matching colors. Dashed gray arrows indicate hashes to the phantom entry. For example, since the filter contains* y, *the bitwise-XOR* $101 \oplus 000 \oplus 110 \oplus 000$ *is the fingerprint* 011. *See Figure 7 and Figure 8 for the induced system of equations and hypergraph.*

Surprisingly, only the conditional branch algorithm achieved higher query throughput than the 4-uniform binary fuse filter, and even this fell short of an acceptable time-space tradeoff. Since we expected nearly all cycles to be spent stalled on memory loads, reducing cache misses per query should have improved throughput proportionally. In practice, however, only 80% of cycles were stalled on loads, and there were no other stalls.

We hypothesize that out-of-order (OoO) execution stalls before it can compute load addresses for the next key. As a result, when outstanding loads complete, hashes for the next query have not yet been computed, and the subsequent loads are delayed by the full hash latency. This dependency also explains the conditional branch algorithm's advantage: its lower hash latency reduces this delay.

The root cause is likely one of two structural hazards: either a shortage of miss status holding registers (MSHRs) prevents prefetching the next key (and therefore precomputing its hashes), or the reorder buffer (ROB) is too small to precompute enough hashes for future keys. Based on microarchitectural parameters and the benchmark assembly, the first explanation is more likely.

2.2.3 Future Work

We see four promising directions for future work. First, batching may yield a non-uniform binary fuse filter with a competitive time-space tradeoff. Second, interleaving the segments of a binary fuse filter could improve cache locality during queries. Third, introducing additional weak points into the fuse graph would enable greater parallelism during filter construction. Currently, the fuse graph has only two weak points, one at each end, limiting peeling to two threads. The

recommended workaround is sharding, but this may sacrifice memory efficiency. Fourth, we aim to prove the conjectures stated in Section 2.2.1, for which the methods of [3] and [9] may be useful.

2.3 Reflection

This project strengthen my ability to enter a new subfield independently through a literature review. Reading many approximate membership query papers, from theoretical analysis to engineered implementations, gave me enough grounding to pose original research questions. One area I want to develop further is posing robust questions, where even a negative result is informative. If non-uniform filter's microarchitectural bottleneck cannot be overcome, that facet is a total loss.

The most demanding aspect of this project was maintaining multiple representations of the same structure simultaneously: a filter is a data structure, a system of linear equations, and a hypergraph. Initially, I found it challenging to move fluidly between these views. Over time this became natural, and I applied it most directly when using cluster resources to empirically explore non-uniform hypergraphs, and translate the result into both formal conjectures and a concrete implementation.

The performance engineering work expanded my understanding of modern processor architecture. Attempting to make the non-uniform filter practical required reasoning carefully about superscalar out-of-order execution and non-blocking caches, interpret hardware performance counters, and diagnose microarchitectural bottlenecks.

Finally, I want to thank Morgan, Tom, Vincent and Dan. Collaborating with them is definitely a highlight of my academic experience.

3 Tuning Secure Learned Bloom Filters

This project was completed for CSC 592: Machine Learning Security.

3.1 Background

Database systems often require fast and small set data structures. For example, RocksDB [4] reduces IO operations by filtering queries before reading from disk. As false positives do not impact correctness, they are permitted with bounded probability. Such approximate membership queries can be much faster and more memory efficient than exact membership queries. However, adversaries may degrade performance by generating false positives.

A Bloom filter [1] is a probabilistic data structure for approximate membership queries. Membership information is stored in an array of bits. To query the

filter, a key is hashed to several entries in the array. If all entries are set to 1 (i.e. the bitwise-AND of the entries is 1), then the query *might* be in the set. False positives occur with probability ϵ. Otherwise, the query *definitely* is not in the set.

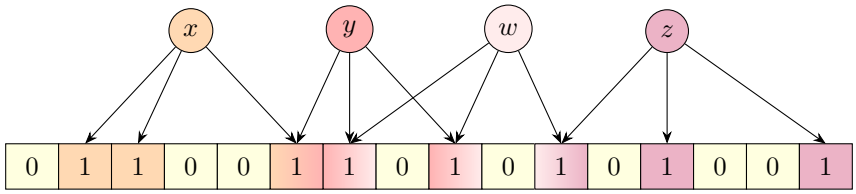

Figure 6: *Bloom filter with 16-entry array and 3 hash functions. The filter contains keys* x, y, w *and* z. *Corresponding entries are marked by arrows and matching colors. For example, since the filter contains* x, *the bitwise-AND* 1 & 1 & 1 *is* 1.

Secure Bloom Filters[5] (SBFs) [**<empty citation>**] extend Bloom filters to provide strong security guarantees. They apply a pseudorandom permutation[6] to keys before querying the filter. If $\mathbf{P} \neq \mathbf{NP}$, there is no probabilistic polynomial time adversary that non-negligibly degrades the false positive rate.

Learned Bloom filters [7] extend Bloom filters with a machine learning model. To query the filter, the machine learning model makes a prediction. If the model predicts the query is in the set, the filter outputs that the key is in the set. This has false positive rate equal to the model. Since the model also makes false negative predictions, a backup Bloom filter containing all false negatives is queried when the model output is negative. This has false positive rate equal to the product of the model's true negative rate and the backup filter's false positive rate.

As strong attacks against machine learning models are known, the false positive rate of a learned Bloom filter can be arbitrarily degraded by an attacker. Secure Learned Bloom Filters prevent this by replacing the backup Bloom filter with a secure Bloom filter and adding a second secure backup Bloom filter to check queries predicted as positive against the set of true positives.

Achieving optimal performance from LBFs requires tuning several parameters including a decision threshold τ, and backup filter false positive rate(s) to attain the desired false positive rate ϵ. Parameter tuning also impacts the filter's memory footprint and, for DBFs, worse-case positive rate. Tuning procedures for LBFs do not directly translate to DBFs, as they do not balance the tradeoff between expected and worst-case false positive rates. Therefore, I ask:

1. How can the decision threshold and backup filter false positive rates be chosen to minimize SLBF memory footprint given desired expected and maximum false positive rates?

[5] Secure Bloom Filters are often (opaquely) called Naor-Eylon filters.
[6] Pseudorandom permutations are not known to exist but are strongly suspected to.

2. How well do SLBFs perform in practice with and without parameter tuning?

3.2 Summary

3.2.1 Tuning Algorithm

We prove that given a fixed decision threshold and desired expected false positive rate ϵ, the secure learned bloom filter size is minimized by taking

$$\mathrm{TP}_{\mathrm{FPR}} = \epsilon \frac{\mathrm{M}_{\mathrm{TPR}}}{\mathrm{M}_{\mathrm{FPR}}} \qquad \mathrm{FN}_{\mathrm{FPR}} = \epsilon \frac{\mathrm{M}_{\mathrm{FNR}}}{\mathrm{M}_{\mathrm{TNR}}}$$

where $\mathrm{TP}_{\mathrm{FPR}}$ is the true positive filter false positive rate, $\mathrm{FN}_{\mathrm{FPR}}$ is the false negative filter false positive rate, and $\mathrm{M}_{\mathrm{TPR}}$, $\mathrm{M}_{\mathrm{FPR}}$, $\mathrm{M}_{\mathrm{FNR}}$ and $\mathrm{M}_{\mathrm{TNR}}$ are the model true positive, false positive, false negative, and true negative rates.

We further prove that the optimal choice of decision threshold minimizes

$$-\mathrm{M}_{\mathrm{TPR}} \ln\left(\frac{\mathrm{M}_{\mathrm{TPR}}}{\mathrm{M}_{\mathrm{FPR}}}\right) - \mathrm{M}_{\mathrm{FNR}} \ln\left(\frac{\mathrm{M}_{\mathrm{FNR}}}{\mathrm{M}_{\mathrm{TNR}}}\right)$$

and provide a $\Theta(n \lg n)$ dynamic programming algorithm to find the optimal threshold. To maintain worst case security guarantees, i.e., a maximum false positive rate, we restrict the choice of τ such that

$$\max\left\{\epsilon \frac{\mathrm{M}_{\mathrm{TPR}}}{\mathrm{M}_{\mathrm{FPR}}}, \epsilon \frac{\mathrm{M}_{\mathrm{FNR}}}{\mathrm{M}_{\mathrm{TNR}}}\right\} \leq \epsilon_{\max}$$

That is, even when the model is forced to only predict positives or only negatives, the false positive rate remains bounded and known.

Here is one sample procedure to optimize the cost function.

1. Sort the learned model confidence scores A on the training set from least to greatest.
2. For each index i,
   (a) Compute the number of elements with negative class in $A[: i]$ and with positive class in $A[i :]$ with a rolling sum.
   (b) Compute the false/true negative/positive rates of the model given that $A[: i]$ is classified as negative and $A[i :]$ is classified as positive.
   (c) Verify that the optimal backup filter false positive rates are less than $\epsilon_{\max}$.
   (d) Track the index with the lowest cost i^*.
3. Output the average between $A[i^* - 1]$ and $A[i^*]$.

3.2.2 Tuned Performance

We construct a SBF and an SLBF on the TODO dataset. We find that for the same memory budget and maximum false positive rate, the SLBF reduces the expected false positive rate from 9.67% to 4.59%, or a 2.1x improvement.

We attack the SLBF with an adaptive black box attack and a targeted random attack. Neither attack is able to degrade the secure learned bloom filter false positive rate worse than $\epsilon_{\max}$.

TODO: A 400–500-word summary of the technical contributions of the work, what was accomplished overall, and the limitations of their work.

3.3 Reflection

TODO: A 250-word self reflection of what skills were developed or enhanced in completing those projects.

4 Entropy-Aware Hashing

This project was completed for CSC 519: Computer Networks.

4.1 Abstract

TODO: A 400–500-word summary of the technical contributions of the work, what was accomplished overall, and the limitations of their work.

4.2 Reflection

TODO: A 250-word self reflection of what skills were developed or enhanced in completing those projects.

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A Non-Uniform Binary Fuse Filters

A.1 Non-Uniform Binary Fuse Filter Alternative Views

$$A[1] \oplus A[3] \oplus A[5] \oplus A[7] = 100$$

$$A[3] \oplus A[5] \oplus A[8] \oplus A[0] = 011$$

$$A[8] \oplus A[9] \oplus A[13] \oplus A[14] = 110$$

$$A[10] \oplus A[13] \oplus A[15] \oplus A[0] = 111$$

Figure 7: *System of linear equations with variables in $\mathbb{F}_2^3$ induced by Figure 5. Each $A[i]$ is the i-th entry in the backing array A (zero-indexed). The keys are x, y, w and z with fingerprints 100, 011, 110 and 111, respectively. Array entries and fingerprints are marked with corresponding key color(s). The phantom entry is colored gray. There is one equation for each key.*

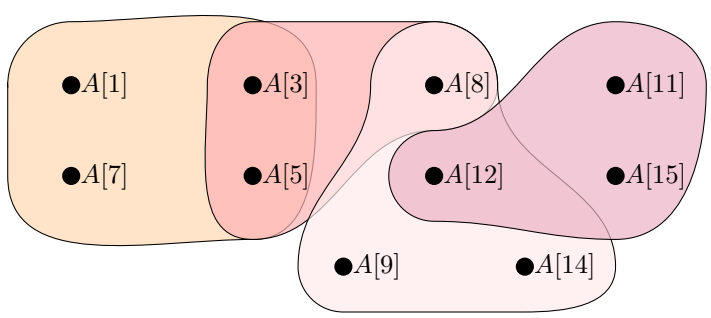

Figure 8: *Hypergraph model of Figure 7. Each $A[i]$ is the i-th entry in the backing array A (zero-indexed). Hyperedges correspond to keys x, y, w and z, colored accordingly. Vertices of degree zero and phantom entry $A[0]$ not shown.*

A.2 Non-Uniform Binary Fuse Filter Query Algorithms

Algorithm 1 Conditional branch algorithm.

```
function Contains(k)
    f ← Fingerprint(k)
    f′ ← A[Hash1(k)] ⊕ A[Hash2(k)] ⊕ A[Hash3(k)]
    if IsFourEdge(k) then
        f′ ← f′ ⊕ A[Hash4(k)]
    end if
    return f == f′
end function
```

Algorithm 2 Branchless algorithm.

```
function Contains(k)
    f ← Fingerprint(k)
    f′ ← A[Hash1(k)] ⊕ A[Hash2(k)] ⊕ A[Hash3(k)]
    f′ ← f′ ⊕ (-IsFourEdge(k) & A[Hash4(k)])
    return f == f′
end function
```

Algorithm 3 Phantom entry algorithm.

```
function Contains(k)
    f ← Fingerprint(k)
    f′ ← A[Hash1(k) + 1] ⊕ A[Hash2(k) + 1] ⊕ A[Hash3(k) + 1]
    f′ ← f′ ⊕ A[-IsFourEdge(k) & (Hash4(k) + 1)]
    return f == f′
end function
```

B Tuning Secure Learned Bloom Filters

C Entropy-Aware Hashing